

SENIOR CAPSTONE PROJECT

SpotMini

Autonomous Robotic Deployment Platform

Team: Diviprakash Kuppusamy, Dhruv Nasit
Built at Penn State University · College of Engineering

Sound familiar?

Your team faces real risks every day — and existing tools aren't solving it.



Dangerous environments

Sending people into hazardous sites — unstable structures, chemical concerns, confined spaces — puts workers at risk every time.



Slow, labor-intensive inspections

Manual walkthroughs take hours. Scheduling is hard. Coverage is inconsistent. Important issues get missed.



Inconsistent data

Reports depend on whoever did the walkthrough that day. You never get the same level of detail twice.

What is SpotMini?

A four-legged robot that goes where your team can't — or shouldn't have to.

SpotMini is a quadrupedal (four-legged) robot built to navigate real-world environments: stairs, debris, uneven ground, tight corridors. It carries sensors, streams live video, and collects data — all while being controlled through a simple web dashboard that anyone on your team can use.

No robotics degree. No coding. Just results.



Moves through any terrain



Streams live video & sensor data



Controlled from any web browser

How it works

Three steps. Zero technical experience required.

1



Set your mission

Log into the SpotMini web dashboard. Type in your starting point, set waypoints on the map, and choose what data you want to collect. Drag, drop, done.

2



Deploy SpotMini

Hit go. SpotMini navigates the environment autonomously, avoids obstacles, keeps its balance on any terrain, and streams live video back to your screen.

3



Get your results

Receive a full report: video footage, sensor readings, flagged anomalies, and a timestamped log — ready to share with your team immediately.

Everything your team needs

One platform. Every tool included.



Live Dashboard

Real-time robot status, battery, camera feeds, and mission progress in one browser tab.



Remote Control

Take manual control any time — keyboard, gamepad, or touchscreen. Full 360° view.



Mission Planner

Build inspection routes visually. Schedule recurring patrols or one-off missions in seconds.



Fleet Management

Running more than one robot? Monitor your whole fleet — status, tasks, and health — from one screen.



Instant Alerts

Get notified the moment SpotMini detects something unusual — no need to watch the feed constantly.



Full Reports

Every mission generates a complete report with video, sensor data, and a timestamped activity log.

Built for your industry

SpotMini is already solving problems across these sectors.



Industrial Facilities

The problem:

Routine equipment checks are costly and slow.

SpotMini:

SpotMini automates walkthroughs and flags anomalies instantly.



Construction Sites

The problem:

Progress tracking requires someone on-site constantly.

SpotMini:

SpotMini documents site conditions daily with no manual effort.



Security & Patrol

The problem:

After-hours perimeter checks depend on staffing.

SpotMini:

SpotMini patrols autonomously 24/7, alerting on anything unusual.



Utilities & Energy

The problem:

Substation and pipeline inspections are high-risk.

SpotMini:

SpotMini enters hazardous zones so your team doesn't have to.

Your workflow, transformed

Without SpotMini

- Workers enter hazardous zones manually
- Inspections take hours per walkthrough
- Data quality depends on who's doing it
- Scheduling gaps leave blind spots
- Reports written by hand after the fact

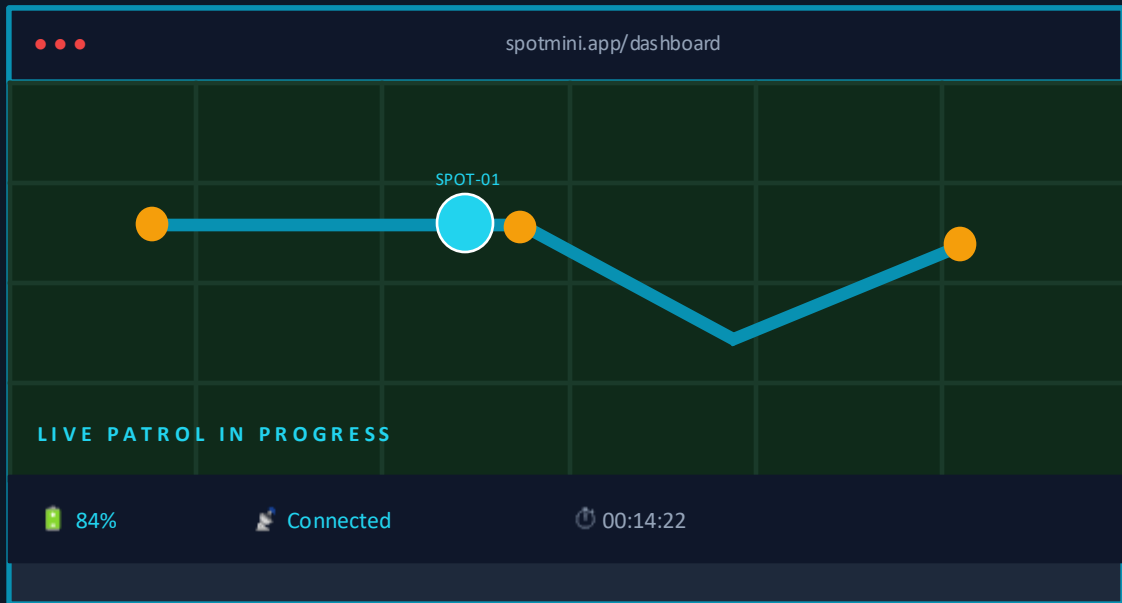
vs

With SpotMini

- Robots enter hazardous zones instead
- Missions run in the background automatically
- Consistent data collected the same way every time
- 24/7 coverage with scheduled patrols
- Full reports generated and delivered automatically

See it in action

The SpotMini dashboard — built for real operations teams, not engineers.



No setup needed — log in and go



Works on any laptop, tablet, or phone



Your team trained in under an hour



All data stored and searchable

Who we are

We're engineers who believe the best technology should be the most accessible.

SpotMini was born as a senior capstone project at Penn State University's College of Engineering. We saw how powerful quadrupedal robots had become — and how frustratingly hard they were for real businesses to actually use.

So we built SpotMini: the robot platform and the software layer that makes it work for any team.



Divi

Robotics & Control Systems

Leads locomotion, gait control, and simulation-to-real transfer.



Dhruv

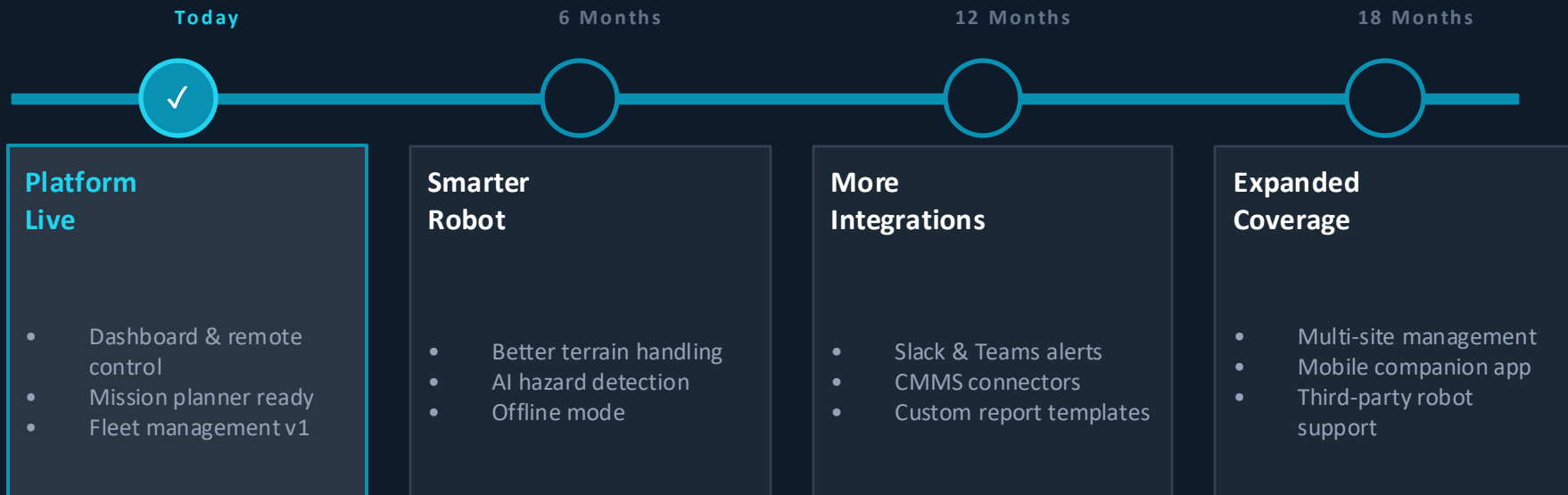
Software & Platform

Leads the web platform, API integrations, and fleet management.



What's coming

SpotMini is growing. Here's what's on the way for you.



Ready to get started?

Getting SpotMini working for your team is easier than you think.

01

Book a free demo

See SpotMini in action live. We'll walk through a real deployment scenario specific to your industry.

02

Pilot in your space

We bring SpotMini to your facility for a hands-on trial. Your team runs it, we handle the setup.

03

Go live

Once you're confident, we deploy, train your team (under an hour), and you're fully operational.

Contact us

Divi

dvk5621@psu.edu

Dhruv

drn5166@psu.edu

spotmini.app

Your team deserves better tools.

SpotMini puts robots to work so your people don't have to take the risk.

Thank You

Divi

dvk5621@psu.edu

Dhruv

drn5166@psu.edu